

Current Electricity

- The rate of flow of charge through any section of wire or conductor is known as current electricity.

It is denoted by i or I

i.e

$$i = \frac{q}{t}$$

or

$$i = \frac{ne}{t}$$

In elementary form.

$$i = \frac{dq}{dt}$$

$$\text{or } i = \frac{dne}{dt}$$

$$i = \left(\frac{dn}{dt} \right) \cdot e$$

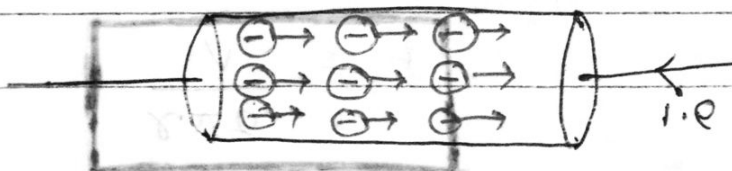
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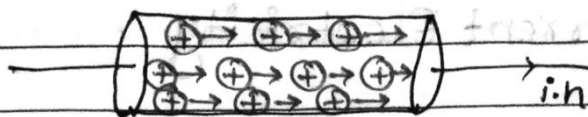
$$i = \left(\text{The rate of flow of no. of free electrons} \right) \times \text{electrons}$$

- Its SI unit and dimension is Ampere (A) and $[A]$ or $[I]$ respectively.

Also, $1A = 1CS^{-1}$

- * Direction of Current :- It flows in the opposite direction of flow of electrons and in the same direction of flow of holes (protons).

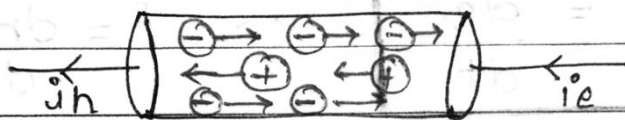




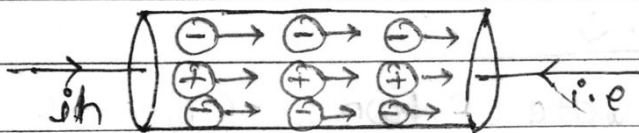
Hence the total current through the conductor is given by.

$$i = |i_e \pm i_h| = \left(\frac{dn_e}{dt} \pm \frac{dn_h}{dt} \right) e$$

+ve and -ve sign is taken as according to electrons and holes are flowing in opposite and same dirn respectively.



$$i = i_e + i_h$$



$$i = |i_e - i_h|$$

*. Current due to revolving charge :-

$$i = \frac{q}{t}$$

$$\Rightarrow$$

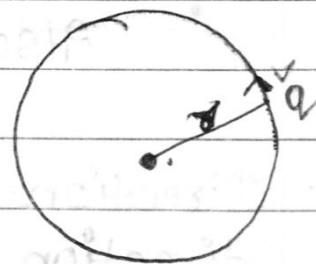
$$\frac{q}{T}$$

$$i = qf$$

$$i = nef$$

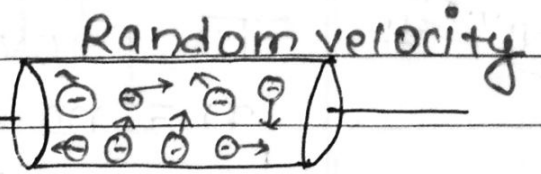
where,

$$f = \frac{v}{2\pi r}$$

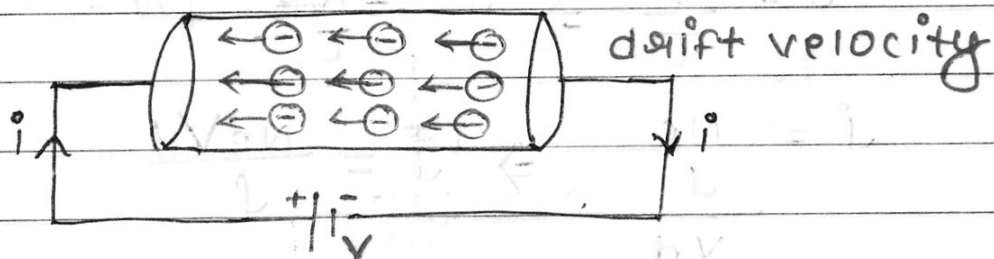


*. Microscopic analysis of current :- In free state of any conductor its free electrons (valence electrons) are moving randomly. This time velocity of electron is known as random velocity.

• The total random velocity of electron become zero. Hence no any current will pass through the conductor.



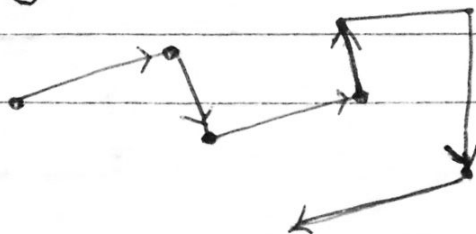
• When potential difference across any conductor then its free electrons are rearranged in a particular direction. Hence net velocity of does not equal to zero.



Hence some current will passing through the conductor.

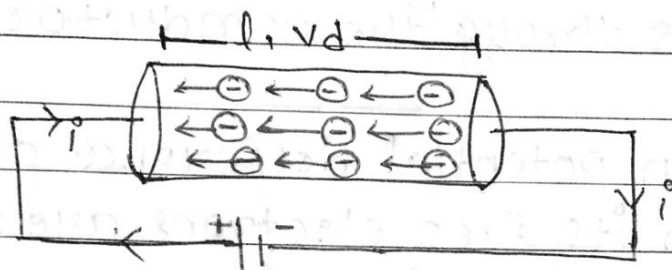
• This time velocity of electron is known as drift velocity.

• After collision of free electron with lattice or +ve ion path the electron is a straight line in zig-zag form.



• Time of relaxation (τ):- The time gap between two successive collision of free electron with +ve ion or lattice is known as time of relaxation.

To prove that $j = nev_d A$ where,
 n = no. of free electron per unit volume
 $e = 1.6 \times 10^{-19} \text{ C}$, v_d = drift velocity
 A = Area of cross section.



Now,

$$j = \frac{q}{t} \Rightarrow j = \frac{Ne}{t} \quad [N = \text{No. of Free electrons}]$$

$$j = \frac{Ne}{l \cdot v_d} \Rightarrow j = \frac{NeV_d}{l}$$

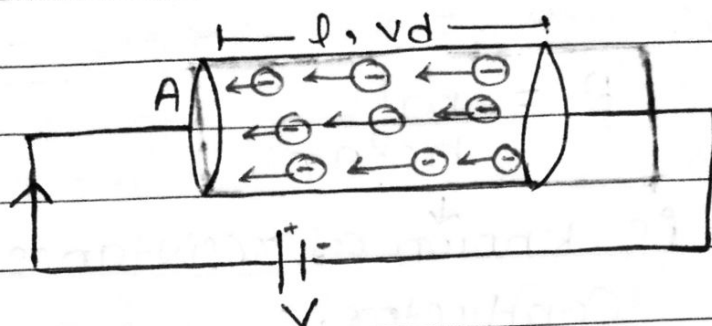
$$j = \frac{NeV_d}{l \cdot A} \cdot A \Rightarrow j = \frac{NeV_d A}{\text{Vol}}$$

$$j = \frac{N}{\text{Vol}} \cdot e v_d A \Rightarrow \boxed{j = n e v_d A}$$

where,

$n = \frac{N}{\text{Vol}}$ = No. of free electron per unit volume.

Using the concept of drift velocity prove that Ohm's law :-



Force upon an electron

$$F = qE$$

$$F = eE$$

$$ma = eE$$

$$a = \frac{eE}{m} \quad \text{--- (1)}$$

Now, $v = u + at$

$$v_d = 0 + a\tau$$

$$v_d = a\tau$$

$$v_d = \frac{eE}{m} \tau$$

$$v_d = \frac{e\tau}{m} \cdot E$$

$$v_d = \frac{e\tau}{m} \cdot \frac{V}{l} \quad \left[E = \frac{V}{l} \right]$$

$$\frac{j}{neA} = \frac{e\tau V}{m l} \quad [j = nev_d A]$$

$$V = \frac{j m l}{ne^2 A \tau}$$

$$\text{Or, } V = j \left(\frac{m l}{n e^2 A \tau} \right)$$

$$\boxed{V = j R}$$

where,

$$\boxed{R = \frac{m l}{n e^2 A \tau}}$$

It is known as resistance of the Conductor.

Hence ohm's law proved

Again,

$$R = \frac{m l}{n e^2 A \tau}$$

$$R = \left(\frac{m}{n e^2 \tau} \right) \frac{l}{A}$$

$$\boxed{R = \rho \frac{l}{A}}$$

where,

$$\rho = \frac{m}{n e^2 \tau}$$

and it is known as resistivity or specific resistance.

- Conductivity :- The reciprocal of resistivity is known as conductivity.

$$\text{i.e. } \boxed{\sigma = \frac{1}{\rho}}$$

Row

• Specific resistance is only depending upon the nature of the material. Hence it remain same.

NOTE-(i) $\begin{matrix} T \uparrow & \rho \downarrow & R \uparrow \\ T \downarrow & \rho \uparrow & R \downarrow \end{matrix}$

for conductor and Semiconductor Resistance is increase or decrease according as its temperature increase or decrease respectively.

(ii)

$$R = \frac{\rho L}{A} \quad ; \quad R \propto L^2$$

$$R = \frac{\rho L}{A \cdot l} \quad ; \quad \Rightarrow R_1 \propto L_1^2 \quad \text{--- (i)}$$

$$R = \frac{\rho L}{A \cdot l} \quad ; \quad \Rightarrow R_2 \propto L_2^2 \quad \text{--- (ii)}$$

$$R = \left(\frac{\rho}{\text{vol}} \right) L^2 \quad ; \quad \boxed{\frac{R_1}{R_2} = \left(\frac{L_1}{L_2} \right)^2}$$

Current Density (J) :- It is defined as the ratio of current to the area of cross section.

i.e. $\boxed{J = \frac{i}{A}}$

$$\boxed{j = JA} \quad \text{or} \quad \boxed{j = \vec{J} \cdot \vec{A}}$$

• Its SI unit and dimension is Am^{-2} and AL^{-2} .

* Mobility :- The ratio of drift velocity to the electric field.

i.e. $\boxed{\mu = \frac{V_d}{E}}$

• Its SI unit = $\frac{\text{ms}^{-1}}{\text{N C}^{-1}} \Rightarrow \boxed{\text{N C}^{-1} \text{ms}^{-1}}$

$\Rightarrow (\text{kgms}^{-2})^{-1} \text{cms}^{-1}$

$\Rightarrow \text{kg}^{-1} \text{m}^{-1} \text{s}^2 \text{cms}^{-1}$

$\Rightarrow \boxed{\text{kg}^{-1} \text{Cs}}$

• Its dimension is $[u] = \underline{\text{m}^{-1} \text{AT}^2}$

9. A material 'B' has twice the specific resistance of 'A'. A circular wire made of 'B' has twice the diameter of a wire made of 'A'. Then for the two wires to have the same resistance, the ratio $\frac{l_A}{l_B}$ of their respective lengths must be,

- (a) $1/2$ (b) $1/4$
 (c) 2 (d) 1

Soln- $\rho_B = 2\rho_A$ — (1)

$d_B = 2d_A$

$2\rho_B = 2(2\rho_A)$

$\rho_B = 2\rho_A$ — (2)

$R_A = R_B$

$\frac{\rho_A \cdot l_A}{\pi r_A^2} = \frac{\rho_B \cdot l_B}{\pi r_B^2}$

$\frac{\rho_A \cdot l_A}{r_A^2} = \frac{2\rho_A \cdot l_B}{(2r_A)^2}$

$\frac{l_A}{r_A^2} = \frac{2l_B}{4r_A^2}$

$l_A = \frac{l_B}{2}$

$2 = \frac{l_B}{l_A}$

$\boxed{\frac{l_B}{l_A} = 2}$ ~~1/2~~

Resistance (R):- The inherent property of a conductor due to which it opposes the current passing through it and directly proportional to the length of conductor and inversely proportional to the area of cross-section. when temperature is constant.

$$\text{i.e. } R \propto l$$

$$\text{and } R \propto \frac{1}{A}$$

Now combining the above we get.

$$R \propto \frac{l}{A}$$

$$R = \rho \frac{l}{A}$$

Its SI unit and dimension are ohm (Ω) and $ML^2T^{-3}A^{-1}$ respectively

$$\text{i.e. Also, } 1\Omega = 1VA^{-1}$$

Conductance (G):- The reciprocal of resistance is known as conductance

$$\text{i.e. } G = \frac{1}{R}$$

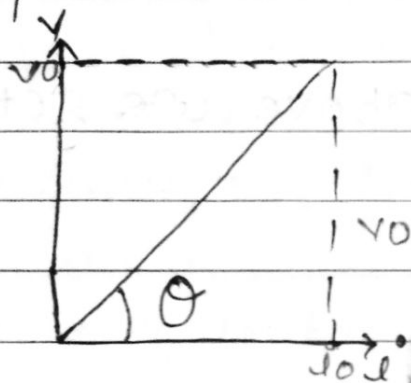
Its SI unit and dimension are 'mho' and $M^{-1}L^{-2}T^3A^{-1}$ respectively

Ohm's law :- It states that if temperature is constant then potential difference across the conductor is directly proportional to current passing through it

i.e. $V \propto i$

$$V = iR$$

• Graph -



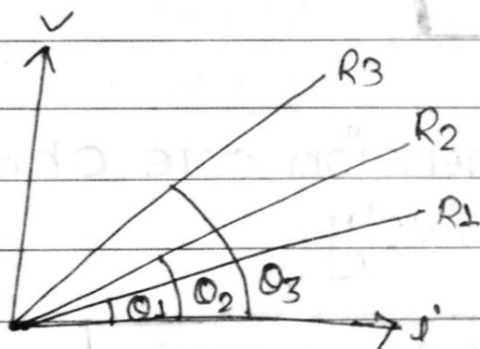
$$\text{Slope} = \tan \theta$$

$$= \frac{V_0}{i_0} = R$$

Hence

$$R = \text{Slope} = \tan \theta$$

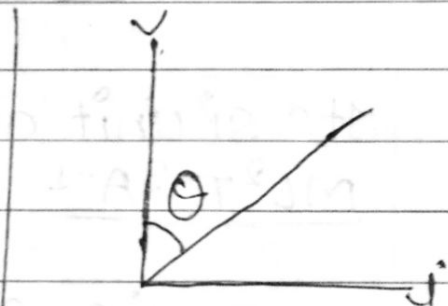
eg (i)



$$\theta_3 > \theta_2 > \theta_1$$

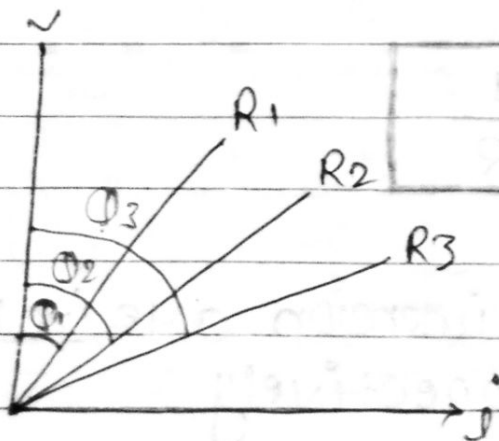
$$\tan \theta_3 > \tan \theta_2 > \tan \theta_1$$

$$R_3 > R_2 > R_1$$



$$R = \frac{1}{\tan \theta} = \frac{1}{\text{Slope}}$$

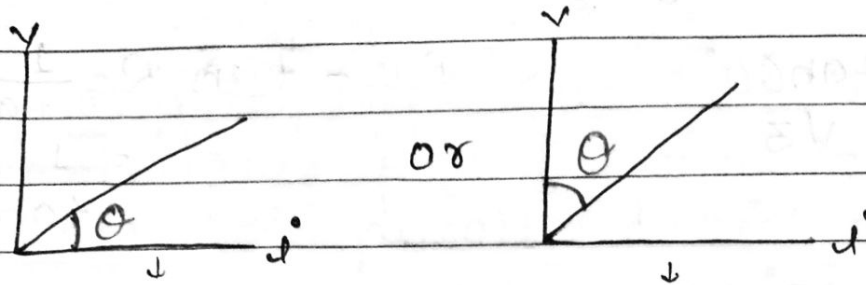
eg (ii)



$$\theta_3 > \theta_2 > \theta_1$$

$$R_3 < R_2 < R_1$$

*. Ohmic Conductor:- The conductor which obey the ohm's law is known as ohmic Conductor.

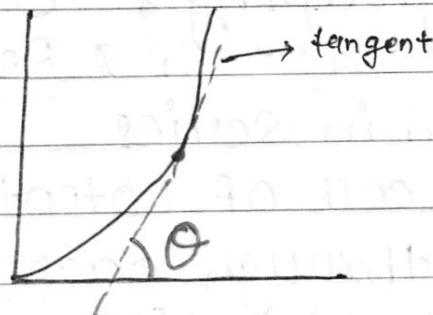


$$R = \tan \theta (\text{slope})$$

or

$$R = \frac{1}{\tan \theta}$$

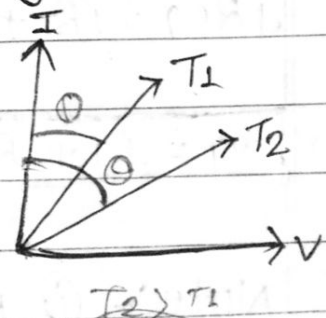
*. Non-Ohmic Conductor:- The conductor which do not obey ohm's law is known as non-Ohmic Conductor.



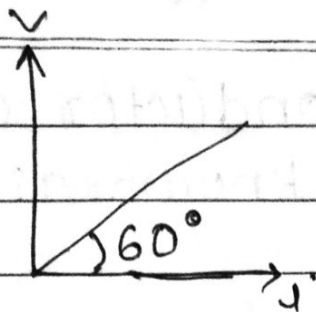
$$R = \tan \theta (\text{slope of tangent})$$

Q1. The current voltage graph for a given metallic Conductor at two different temperatures T_1 and T_2 are as shown in the figure. Then,

- (a) $T_1 > T_2$
- (b) $T_1 = T_2$
- (c) Nothing can be said about T_1 and T_2
- ~~(d) $T_1 < T_2$~~

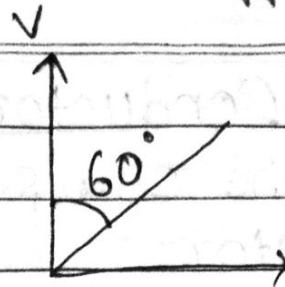


Q2.



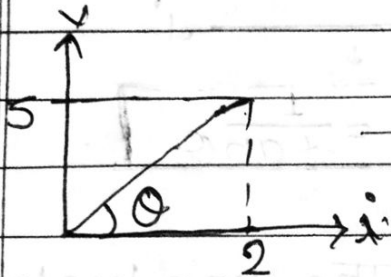
Ans - $R = \tan 60^\circ$
 $= \sqrt{3}$

Q3.



Ans - ~~$R = \tan \theta$~~ $R = \frac{1}{\tan \theta}$
 $= \frac{1}{\tan 60} \Rightarrow \frac{1}{\sqrt{3}}$

Q4.



Sol: - $R = \tan \theta = \frac{P}{b} = \frac{5}{2} \text{ } \Omega$

Grouping of resistance :-

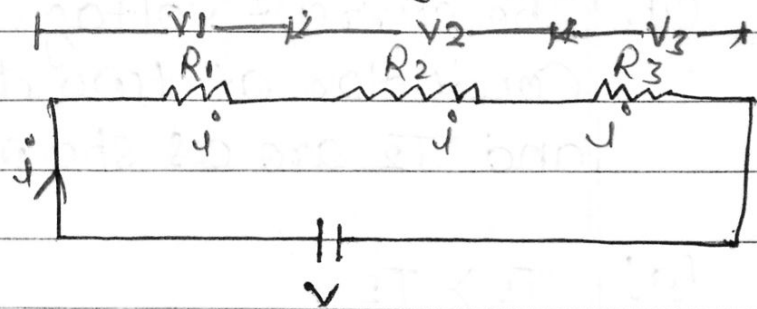
(i) Series grouping :- Consider there are three resistor R_1 & R_2 and R_3 are connected in series.

with a cell of potential V . In this grouping current through each resistor will same and potential difference will be different. Let V_1, V_2 & V_3 are potential difference across R_1, R_2 and R_3 respectively then.

$$V = V_1 + V_2 + V_3$$

$$iR_{eq} = iR_1 + iR_2 + iR_3$$

$$R_{eq} = R_1 + R_2 + R_3$$



Note:- (i) $R_{eq} = R_1 + R_2 + R_3 + \dots + R_n$

(ii) If $R_1 = R_2 = R_3 = \dots = R_n = (R \text{ say})$

then

$$R_{eq} = R + R + R + \dots R_n$$

$$R_{eq} = nR$$

(ii)

Parallel grouping :- Consider there are three resistor R_1 , R_2 and R_3 are connected in parallel. with a cell of potential V . In this grouping current through each resistor will be different and potential difference will be same. Let V_1, V_2, V_3 are potential difference same and current are different i_1, i_2 & i_3 across R_1, R_2 & R_3 respectively then.

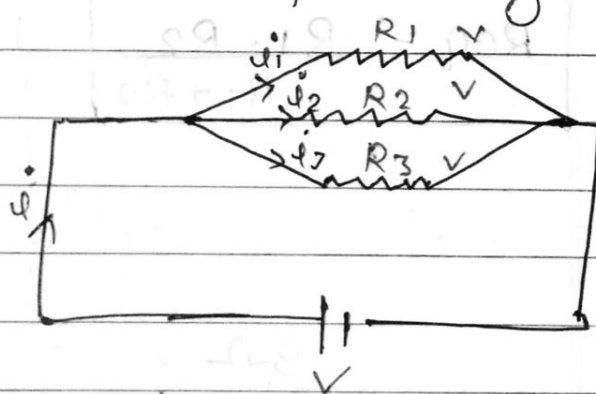
current = different

potential difference = same

$$i = i_1 + i_2 + i_3$$

$$\frac{V}{R_{eq}} = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}$$

$$\frac{V}{R_{eq}} = \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right)$$



$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

NOTE (i) $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_n}$

(ii) If $R_1 = R_2 = R_3 = \dots R_n = R$ (say) then,

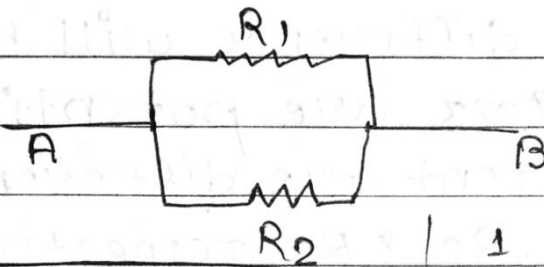
$$\frac{1}{R_{eq}} = \frac{1}{R} + \frac{1}{R} + \frac{1}{R} + \dots + \frac{1}{R_n}$$

$$\frac{1}{R_{eq}} = \frac{n}{R}$$

$$R_{eq} = \frac{R}{n}$$

* Problem :-

(i)



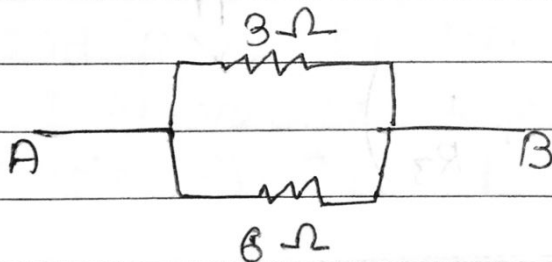
$$R_{eq} = \frac{R_1 \cdot R_2}{R_1 + R_2}$$

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{1}{R_{eq}} = \frac{R_2 + R_1}{R_1 \cdot R_2}$$

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

eg - (i)

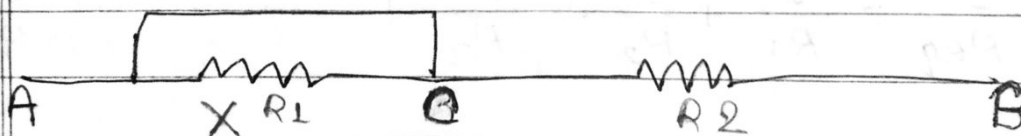


$$R_{eq} = \frac{3 \times 6}{3 + 6} = \frac{18}{9}$$

$$R_{eq} = 2 \Omega$$

Ans.

(ii) Short circuited :-



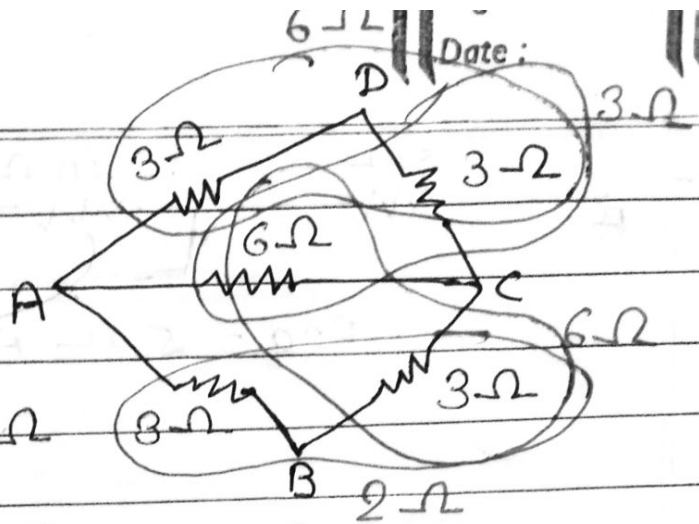
$$R_{eq} = R_2$$

$$Q1. \quad 3\Omega + 3\Omega = 6\Omega$$

$$\Rightarrow \frac{6 \times 6}{6 + 6} = \frac{36}{12} = 3\Omega$$

$$\Rightarrow 3\Omega + 3\Omega \Rightarrow 6\Omega$$

$$\Rightarrow \frac{6 \times 3}{6 + 3} = \frac{18}{9} = 2\Omega$$



$$Q2.2 \quad \frac{1}{R} = \frac{1}{8} + \frac{1}{16} + \frac{1}{16} = \frac{2+1+1}{16}$$

$$= \frac{4}{16} = \frac{1}{4}$$

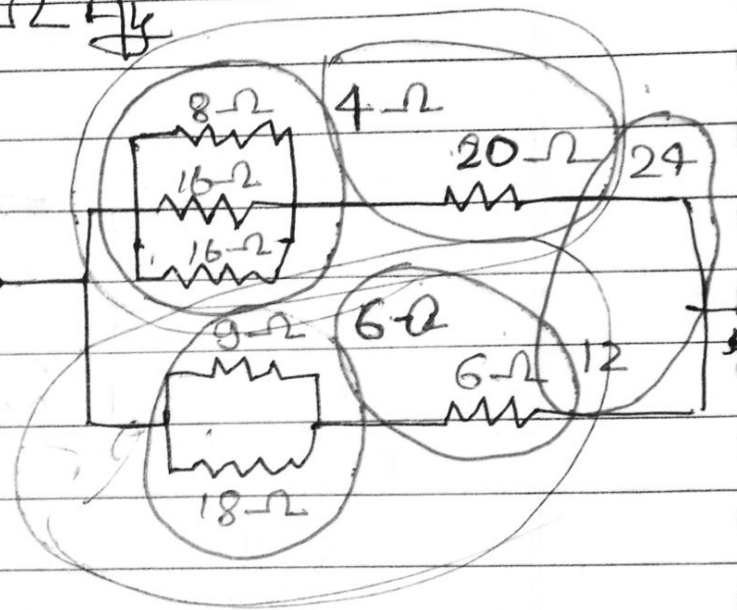
$$R = 4$$

$$R = 20 + 4 \Rightarrow 24$$

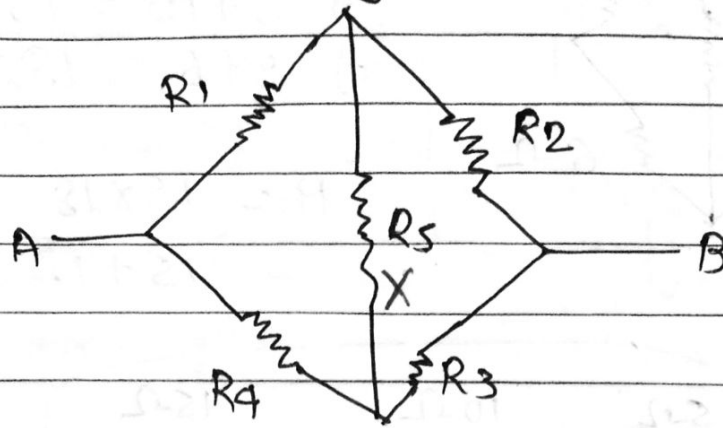
$$\frac{9 \times 18}{9 + 18} = \frac{9 \times 18}{27} = 6\Omega$$

$$R = 6 + 6 \Rightarrow 12\Omega$$

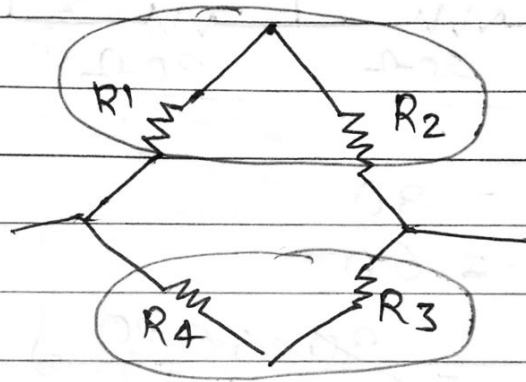
$$\therefore \frac{12 \times 24}{12 + 24} = \frac{12 \times 24}{36} = 8\Omega$$



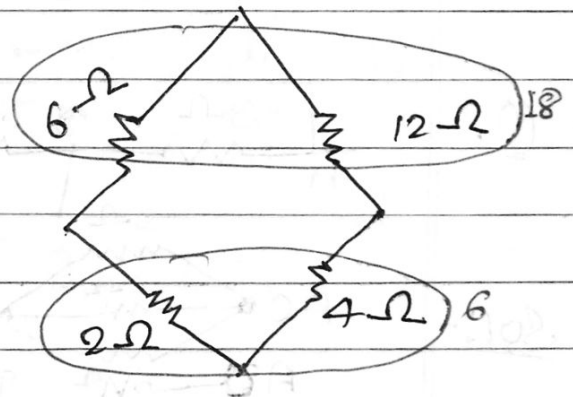
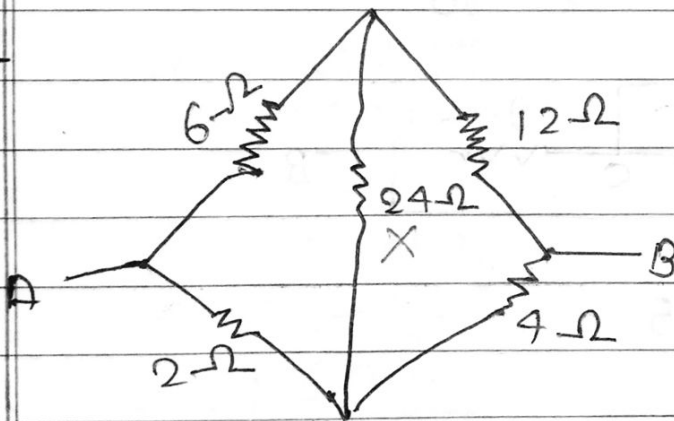
*. Wheat Stone bridge:-



If $R_1 R_3 = R_4 R_2$ then it is known as balanced Wheat stone bridge

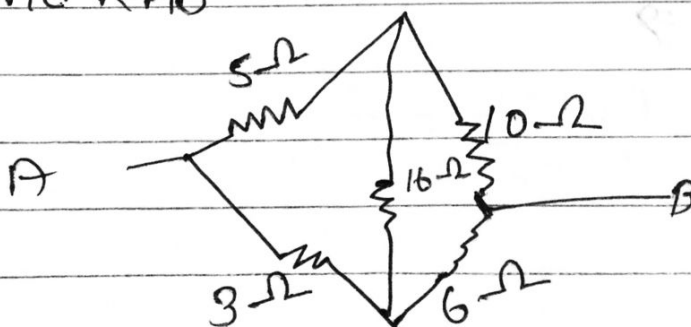


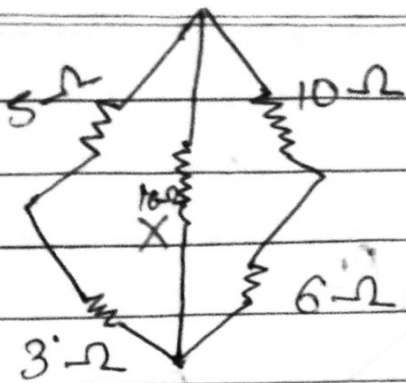
eg:-



$$= \frac{18 \times 6}{18 + 6} \Rightarrow \frac{18 \times 6}{24} = \frac{9}{2} \text{ } \Omega$$

Q. Find R_{AB}



Sol:-

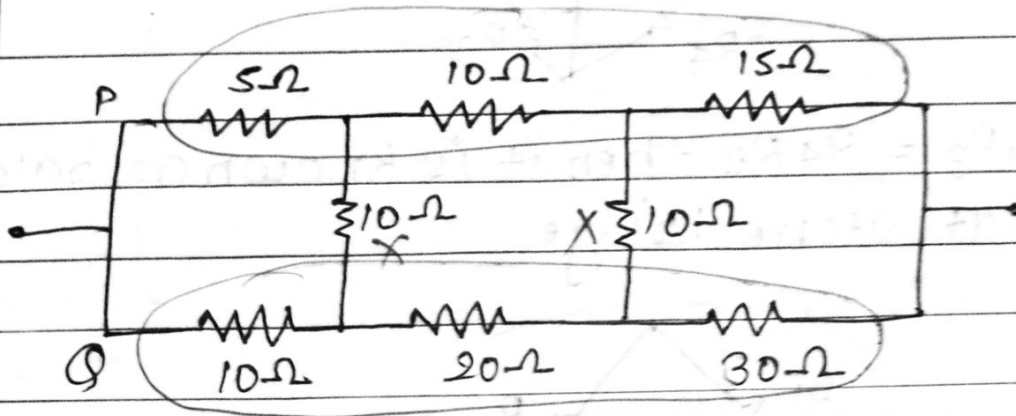
$$\Rightarrow 5 + 10 = 15$$

$$\Rightarrow 3 + 6 = 9$$

$$R = \frac{15 \times 9}{15 + 9}$$

$$= \frac{15 \times 9}{24} = \frac{45}{8} \Omega$$

Q2.

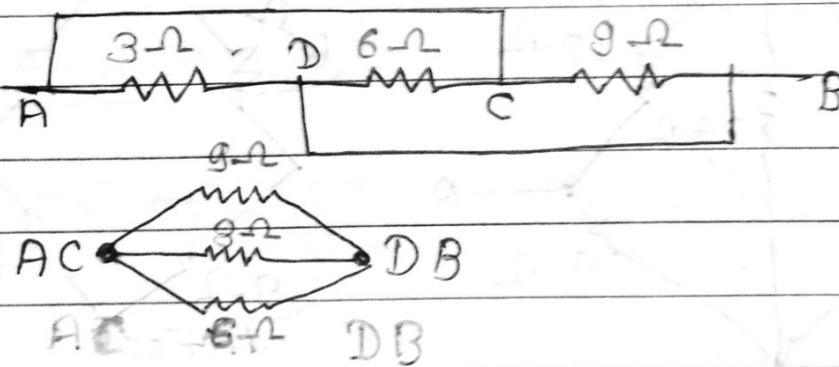


$$R = 5 + 10 + 15 = 30$$

$$R = 10 + 20 + 30 = 60$$

$$R = \frac{30 \times 60}{30 + 60} = \frac{30 \times 60}{90} = 20 \Omega$$

Q.

Sol:-

$$\frac{1}{R} = \frac{1}{6} + \frac{1}{3} + \frac{1}{9} = \frac{3+6+2}{9} = \frac{11}{9}$$

$$\frac{1}{R} = \frac{11}{9} \Rightarrow$$

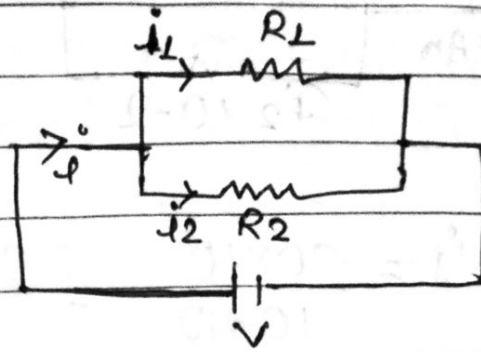
$$R = \frac{9}{11}$$

* Important points -

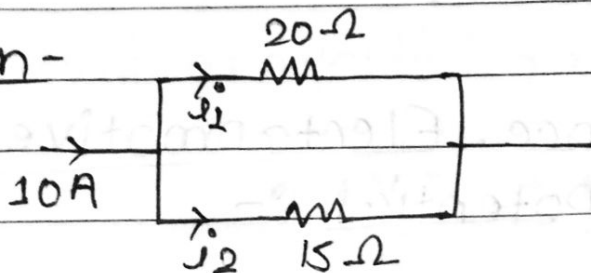
①

$$i_1 = \frac{i \times R_2}{R_1 + R_2}$$

$$i_2 = \frac{i \times R_1}{R_1 + R_2}$$



1 Question -

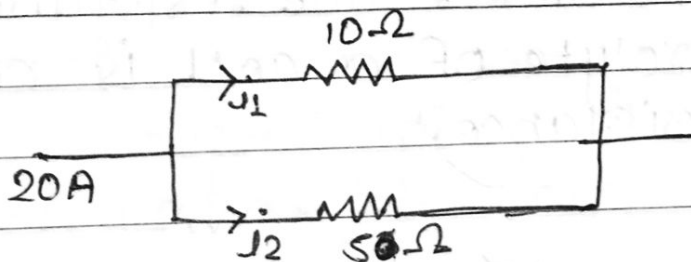


$$\text{Sol: } i_1 = \frac{10 \times 15}{20 + 15} = \frac{10 \times 15}{35}$$

$$= \frac{30}{7} \text{ A}$$

$$i_2 = \frac{10 \times 20}{35} = \frac{40}{7} \text{ A}$$

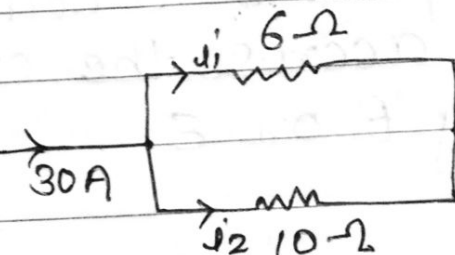
2 Q.



$$\text{Sol: } i_1 = \frac{20 \times 50}{20 + 50} = \frac{20 \times 5}{15} = \frac{20}{3} \text{ A}$$

$$i_2 = \frac{20 \times 10}{15} = \frac{40}{3} \text{ A}$$

3 Q.

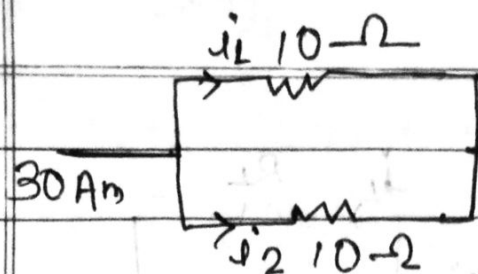


$$\text{Sol: } i_1 = \frac{30 \times 10}{6 + 10}$$

$$= \frac{15 \times 10}{8} = \frac{75}{4}$$

$$i_2 = \frac{30 \times 6}{6 + 10} = \frac{15 \times 6}{8} = \frac{45}{4} \text{ A}$$

Q 4.

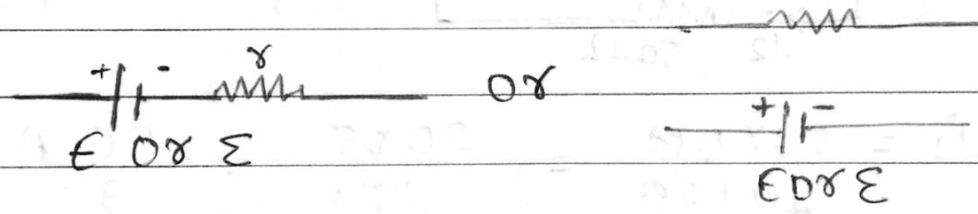


$$i_1 = \frac{30 \times 10}{10 + 10} = \frac{30 \times 10}{20} = 15 \text{ A}$$

$$i_2 = \frac{30 \times 10}{10 + 10} = \frac{30 \times 10}{20} = 15 \text{ A}$$

*. Internal resistance, Electromotive force and Terminal potential :-

- Internal resistance :- The resistance offered by the electrolyte of a cell is called internal resistance.



- Electromotive force (E.M.F) :- The potential difference between the poles of a cell when it does not connected across the external circuit and denoted by E or $\mathcal{E}$.

- Terminal potential (V) :- The potential difference between the poles of a cell when it connected across the external circuit and denoted by V .

- Terminal potential is always less than the e.m.f because some emfs are loosed to avoiding the internal resistance.
- Terminal potential for discharging condition

$$V' = E - i r$$

↓ loss in potential = $i r$

- Terminal potential for charging condition -

$$V = E + i r$$

Expression for internal resistance :-
Consider an external resistor of resistance R Connected with cell of e.m.f E and internal resistance r as shown in fig.

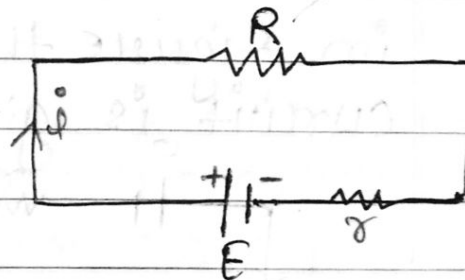
$$R_{eq} = R + r$$

From ohm's law

$$E = i R_{eq}$$

$$E = i (R + r)$$

$$i = \frac{E}{R + r} \quad \text{--- (1)}$$



Now terminal potential across R

$$V = i R$$

$$V = \frac{E}{R + r} \cdot R$$

$$V R + V r = E R$$

$$V \times = ER - V R$$

$$r = \frac{ER - V R}{V}$$

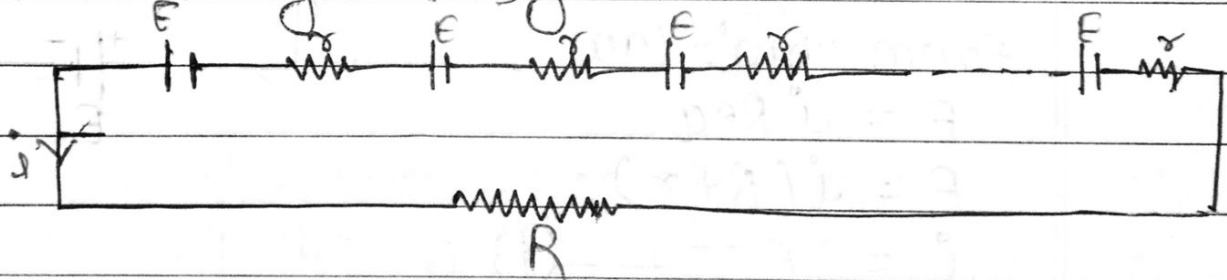
$$r = \frac{ER}{V} - \frac{V R}{V}$$

$$r = \frac{ER}{V} - R$$

$$r = \left(\frac{E}{V} - 1 \right) R$$

*. Grouping of cell :-

- (i) Series grouping :- Consider there are n cells each of emf E and internal resistance r are connected in series with external resistor R . As shown in figure then, current through the circuit is given by.



$$E_{\text{eff}} = E + E + E + \dots \text{to } n$$

$$E_{\text{eff}} = nE$$

$$R_{\text{eq}} = (r + r + r + \dots \text{to } n) + R$$

$$R_{\text{eq}} = nr + R$$

current through circuit is given by

$$i = \frac{E_{\text{eff}}}{R_{\text{eq}}} = \frac{nE}{nr + R}$$

$$i = \frac{nE}{nr + R}$$

If $r \ll R$

If $r \gg R$

$$i = \frac{nE}{R}$$

$$i = \frac{nE}{nr} = \frac{E}{r}$$

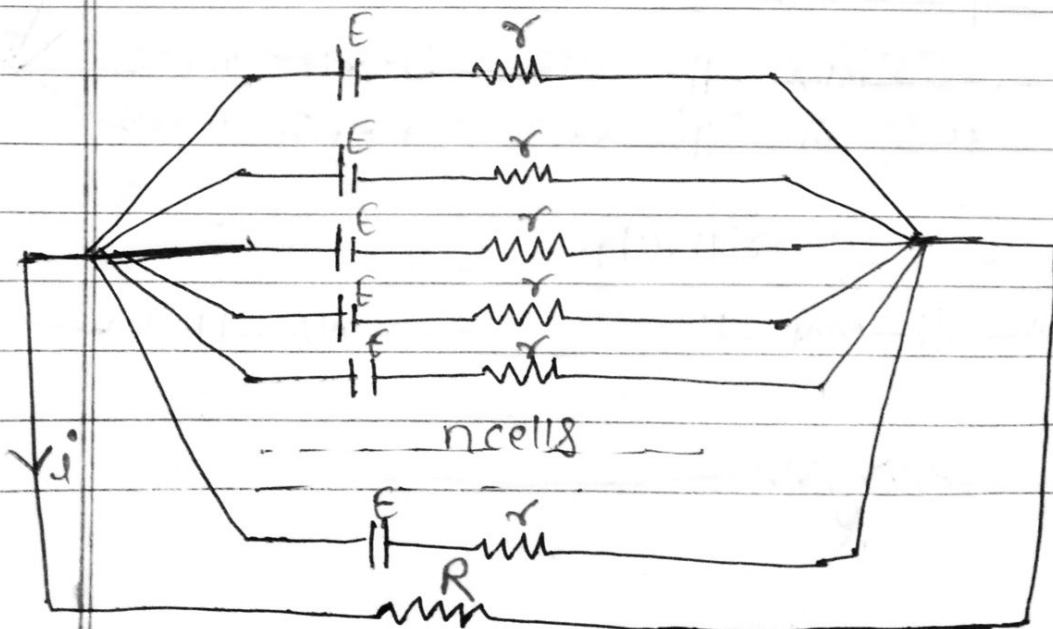
NOTE - If N cells are wrongly connected out of n cells in series grouping then effective e.m.f and current through the circuit is given by

$$E_{\text{eff}} = nE - 2NE$$

$$i = \frac{nE - 2NE}{nr + R}$$

$$i = \frac{(n - 2N)E}{nr + R}$$

(ii) Parallel grouping:- Consider there are n cells each of internal resistance ~~sm~~ r are connected in parallel with a cell of e.m.f E with an external resistor R as shown in Fig.



$$E_{\text{eff}} = E$$

$$R_{\text{eq}} = \frac{\sigma}{n} + R = \frac{\sigma + nR}{n}$$

Now,

$$I = \frac{E_{\text{eff}}}{R_{\text{eq}}} = \frac{E}{\frac{\sigma + nR}{n}} = \frac{nE}{\sigma + nR}$$

$$I = \frac{nE}{\sigma + nR}$$

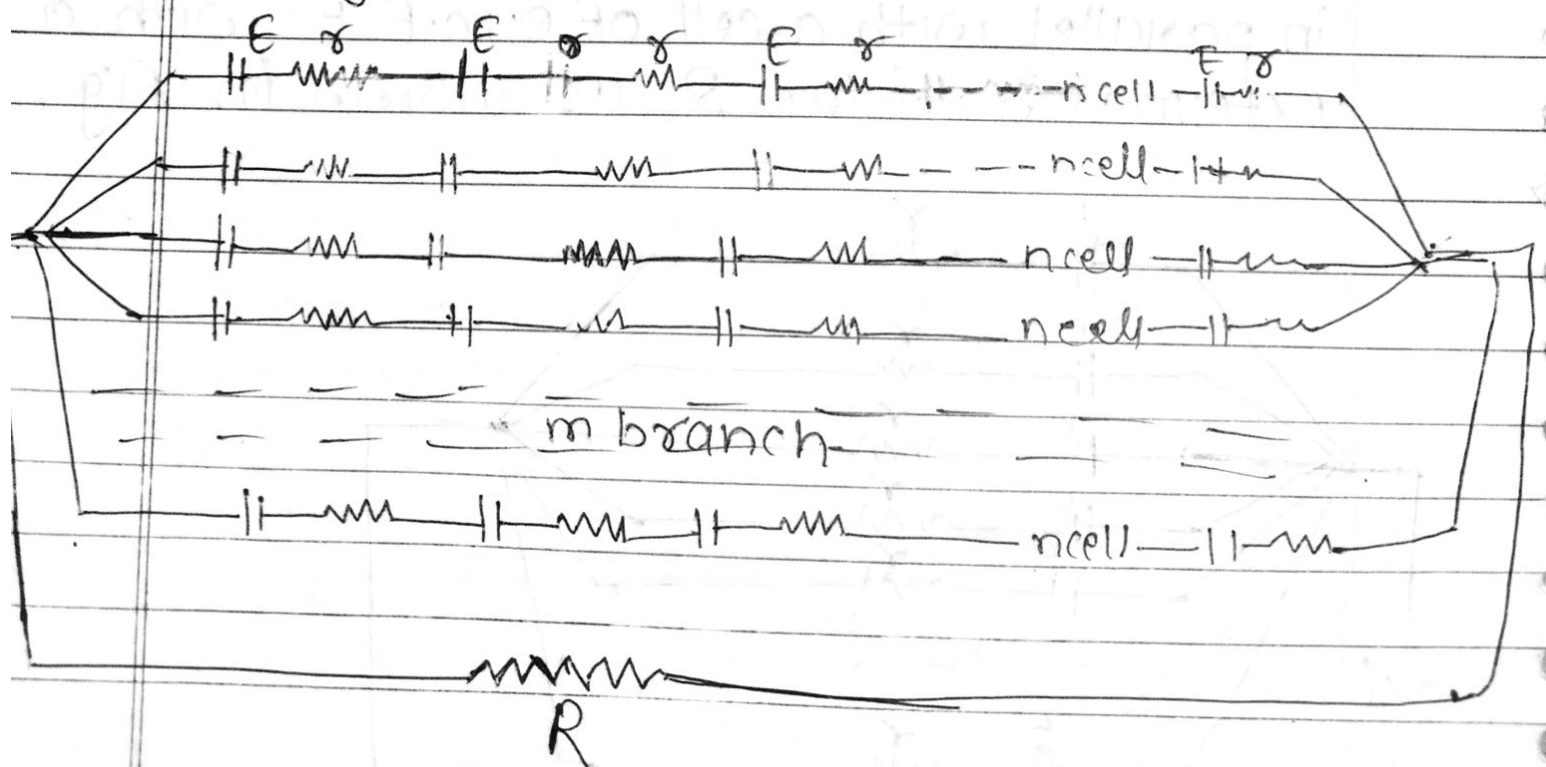
If $\sigma \gg R$ then

$$I = \frac{nE}{R}$$

If $\sigma \ll R$ then

$$I = \frac{E}{R}$$

(iii) Mixed grouping:- Consider there are n cells each of emf E and internal resistance σ are connected in series now, this type of m branch are connected in parallel with an external resistance R as shown in fig.



$$E_{\text{eff}} = nE$$

$$R_{\text{eq}} = \frac{n\gamma}{m} + R = \frac{n\gamma + mR}{m}$$

current through J

$$J = \frac{E_{\text{eff}}}{R_{\text{eq}}} = \frac{nE}{\frac{n\gamma + mR}{m}}$$

$$J = \frac{mnE}{n\gamma + mR} = \frac{NE}{n\gamma + mR}$$

where,

$N = \text{Total no. of cells} = mn$

If $\gamma \ll R$ i.e. $\gamma \approx 0$ then

$$J = \frac{nE}{R}$$

If $\gamma \gg R$ i.e. $R \approx 0$ then

$$J = \frac{mE}{\gamma}$$

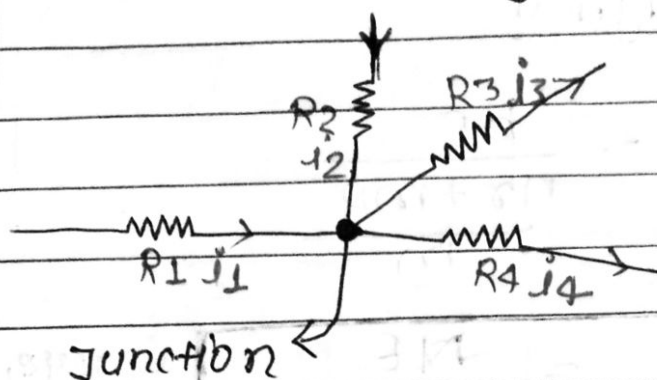
Kirchhoff's Rule :-

- (i) Kirchoff's 1st rule / Junction law :- It states that the algebraic sum of current meeting at a point or at a junction is equal to zero.

i.e.

$$\sum J = 0$$

- Sign Convention - current will be taken as +ve or -ve according to ^{as it incoming} outgoing towards junction or away from junction respectively



incoming = +ve
outgoing = -ve

$$\sum i = 0$$

$$i_1 + i_2 - i_3 - i_4 = 0$$

$$i_1 + i_2 = i_3 + i_4$$

Incoming current = outgoing current

- (ii) Kirchoff's 2nd rule / Loop rule :- It states that in a close circuit the algebraic sum of E.m.f or potential difference is equal to algebraic sum of product of current and resistance.

i.e. $\boxed{\sum E = \sum iR}$

or

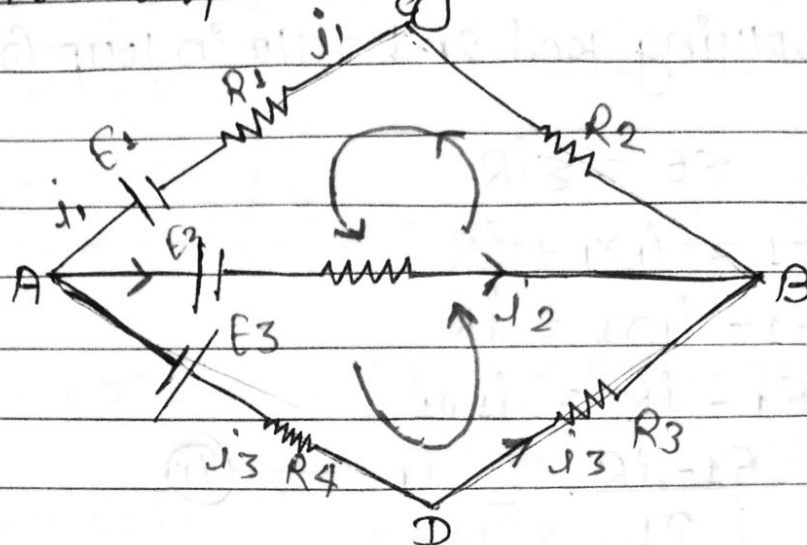
$$\boxed{\sum V = \sum iR}$$

- Sign Convention :-

- (i) Current is taken as +ve or -ve according as it flowing in A.C.W direction or C.W direction respectively.

ii) The emf of a cell is taken +ve or -ve A/c as it send the current in A.C.W or C.W dirⁿ respectively from +ve pole.

eg:-



A.C.W = +ve
C.W = -ve

Applying KVL in ABCA.

$$\sum E = \sum iR$$

$$E_1 + E_2 = i_2 R_5 + i_1 R_1 + i_1 R_4$$

from loop ADBA

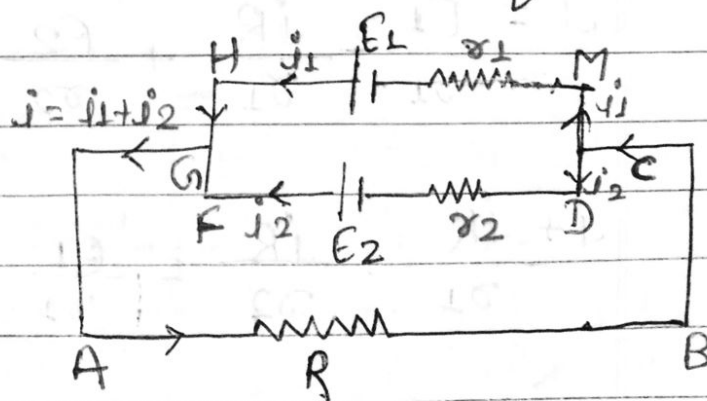
$$-E_2 + E_3 = i_3 R_4 + i_3 R_3 - i_2 R_5$$

#1 Application of Kirchhoff's law :-

Let two cells of e.m.f E_1 & E_2 having internal resistance r_1 & r_2 are connected in parallel an external resistor of resistance R are joined in series as shown in Figure.

Then common potential with R_{eq} is given by.

$$E = \frac{E_1 r_2 + E_2 r_1}{r_1 + r_2}$$



$$\boxed{Reg = \frac{\sigma_1 \sigma_2}{\sigma_1 + \sigma_2} + R}$$

proof:- let $\dot{j} = \dot{j}_1 + \dot{j}_2$ ——— (1)

Applying Kcl 2nd rule in loop GABCMHG

$$\sum E = \sum iR$$

$$E_1 = \dot{j}_1 \sigma_1 + \dot{j} R$$

$$E_1 - \dot{j} R = \dot{j}_1 \sigma_1$$

$$E_1 - \dot{j} R = \dot{j}_1 \sigma_1$$

$$\frac{E_1 - \dot{j} R}{\sigma_1} = \dot{j}_1 \quad \text{--- (i)}$$

for loop GABCDFG

$$\sum E = \sum iR$$

$$E_2 = \dot{j} R + \dot{j}_2 \sigma_2$$

$$\frac{E_2 - \dot{j} R}{\sigma_2} = \dot{j}_2 \quad \text{--- (ii)}$$

Now from eq - (i)

$$\dot{j} = \dot{j}_1 + \dot{j}_2$$

$$\dot{j} = \frac{E_1 - \dot{j} R}{\sigma_1} + \frac{E_2 - \dot{j} R}{\sigma_2}$$

$$\dot{j} = \frac{E_1}{\sigma_1} - \frac{\dot{j} R}{\sigma_1} + \frac{E_2}{\sigma_2} - \frac{\dot{j} R}{\sigma_2}$$

$$\frac{\dot{j} + \dot{j} R}{\sigma_1} + \frac{\dot{j} R}{\sigma_2} = \frac{E_1}{\sigma_1} + \frac{E_2}{\sigma_2}$$

$$j \left(\frac{1+R}{r_1} + \frac{R}{r_2} \right) = \frac{E_1 r_2 + E_2 r_1}{r_1 r_2}$$

$$j \left[\frac{r_1 \cdot r_2 + R r_2 + R r_1}{r_1 r_2} \right] = \frac{E_1 r_2 + E_2 r_1}{r_1 r_2}$$

$$j [r_1 \cdot r_2 + R(r_1 + r_2)] = E_1 r_2 + E_2 r_1$$

$$E_1 r_2 + E_2 r_1 = j (r_1 r_2 + R(r_1 + r_2))$$

$$\frac{E_1 r_2 + E_2 r_1}{r_1 + r_2} = j \left(\frac{r_1 r_2}{r_1 + r_2} + R \right)$$

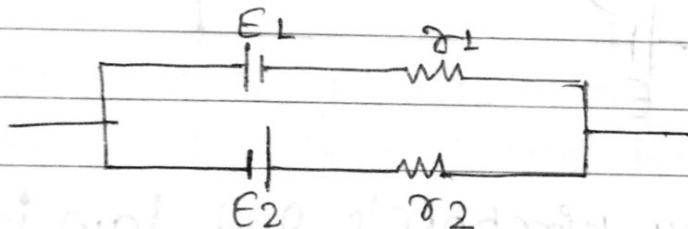
$$E = j R_{eq}$$

$$E = \frac{E_1 r_2 + E_2 r_1}{r_1 + r_2}$$

$$R_{eq} = R + \frac{r_1 \cdot r_2}{r_1 + r_2}$$

proved

NOTE ①



$$E = \frac{E_1 r_2 - E_2 r_1}{r_1 + r_2}$$

②

we know that

$$E = \frac{E_1 r_2 + E_2 r_1}{r_1 + r_2}$$

$$E = \frac{E_1}{r_1} + \frac{E_2}{r_2} + \frac{E_3}{r_3} + \dots$$

$$\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} + \dots$$

$$E = \frac{E_1}{r_1} + \frac{E_2}{r_2}$$

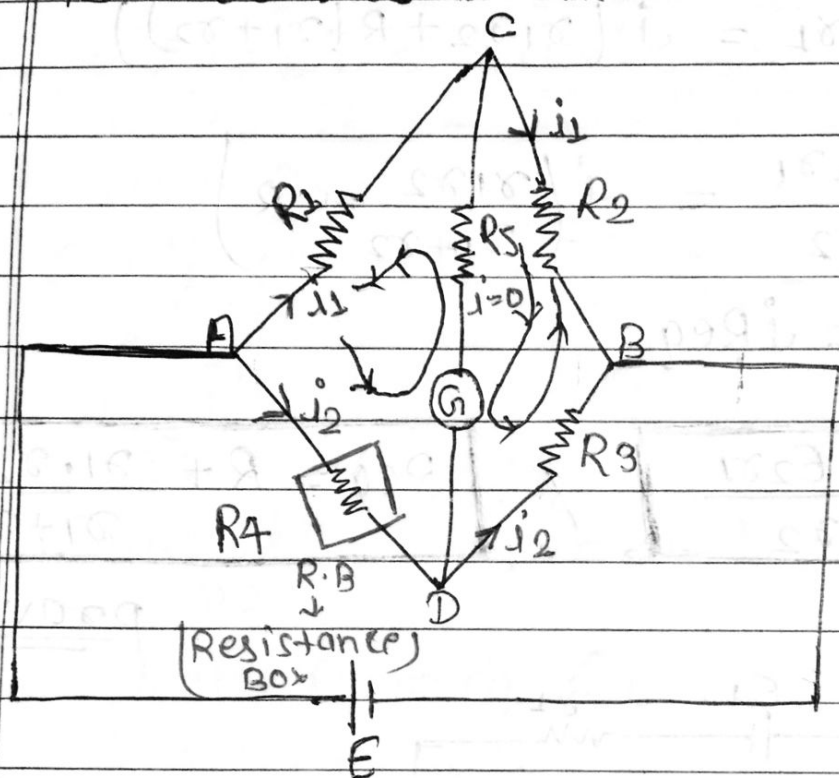
$$\frac{1}{r_1} + \frac{1}{r_2}$$

for 2 cell

②

Wheat Stone bridge :- It is a device which is used to find unknown resistance.

It consists of four resistance R_1, R_2, R_3 & R_4 in four arms and R_5 connected with a galvanometer. R_4 is adjusted in such a way that no current is passing through the galvanometer. This time wheat stone bridge is known as balance wheat stone bridge.



Applying Kirchhoff's 2nd Law in loop ADCA we get

$$\sum E = \sum iR$$

$$0 = i_2 R_4 + 0 \cdot R_5 - i_1 R_1$$

$$\Rightarrow i_1 R_1 = i_2 R_4 \quad \text{--- (1)}$$

Similarly in the loop CDBC

$$0 = 0 + i_2 R_3 - i_1 R_2$$

$$i_1 R_2 = i_2 R_3 \quad \text{--- (1)}$$

From (i) & (ii)

$$\frac{i_1 R_1}{i_1 R_2} = \frac{i_2 R_4}{i_2 R_3}$$

$$\frac{R_1}{R_2} = \frac{R_4}{R_3} \quad \text{or} \quad \frac{R_1}{R_4} = \frac{R_2}{R_3}$$

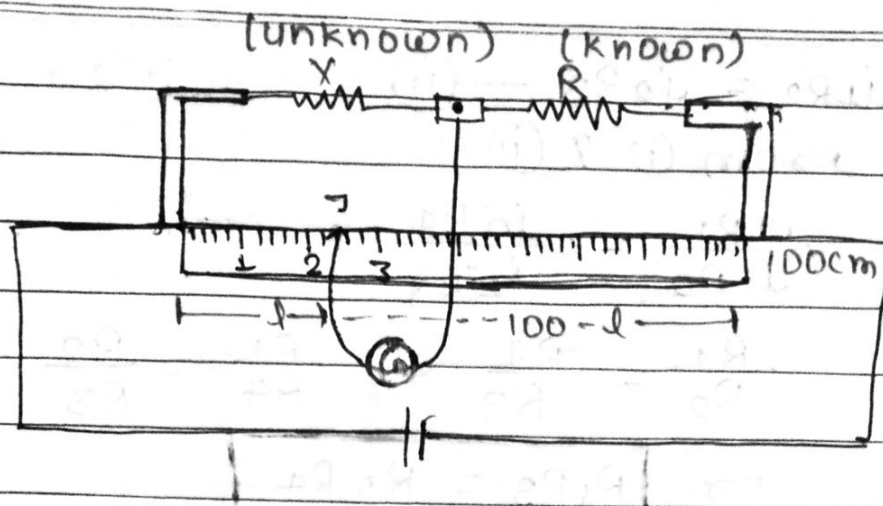
$$\text{or} \quad \boxed{R_1 R_3 = R_2 R_4}$$

→ This is required condition for wheat stone bridge

NOTE - (i) First Kirchhoff's law gives conservation of charge while Second law gives conservation of energy.

Metre bridge / Slide bridge :- It is a device which used to finding out the unknown resistance.

- Principle :- It is working on the principle of balanced wheat stone bridge.
- Construction :- It consist of a metre scale of length 100cm from the both the ends it connected with an unknown resistance (X) and known resistance (R). Between these two resistors a Galvanometer with jockey connected through to find out the null point by sliding on meter scale As shown in figure.



Theory:- Let null point be found by jockey at a distance l cm then form the balance wheat stone bridge.

$$X(100-l) = Rl$$

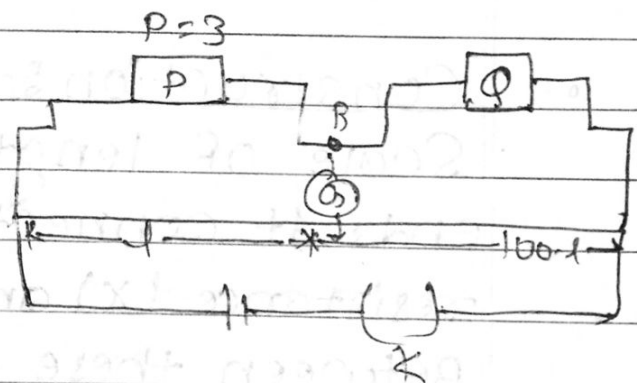
$$\boxed{\frac{X}{R} = \frac{l}{100-l}}$$

Q. In a metre bridge experiment, resistance are connected as shown in figure. The balancing length l_1 is 55 cm now an unknown resistance x is connected in series with p and the new balancing point is found to be 75 cm, The value of x is

(a) $\frac{54}{12} \Omega$ (b) $\frac{20}{11} \Omega$

(c) $\frac{48}{11} \Omega$ (d) $\frac{11}{48} \Omega$

(e) 5Ω



$$3 \times 45 = Q \times 55$$

$$Q = \frac{27}{11}$$

$$(3+x) 25 = Q \times 75$$

$$3+x = \frac{27}{11} \times 3$$

$$3+x = \frac{81}{11}$$

$$x = \frac{81}{11} - 3 = \frac{81-33}{11} = \frac{48}{11}$$

Q2. The resistance in left gap and right gap of a metre bridge are 20Ω and 30Ω respectively. When the resistance in the left gap is reduced to half its value, the balance point shifts by

- (a) 15cm to the right (b) 15cm to the left
(c) 20cm to the right (d) 20cm to the left

Sol: $20(100-l) = 30(l)$

$$200 - 2l = 3l$$

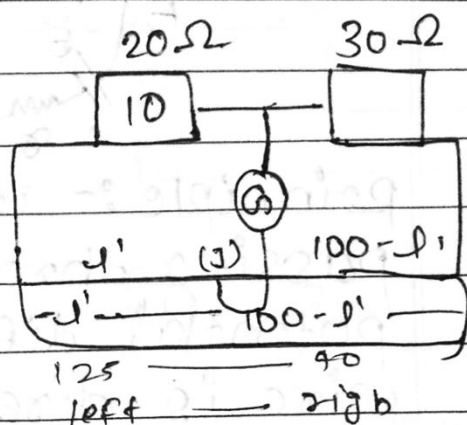
$$l = \frac{200}{5} = 40\text{cm}$$

$$10(100-l') = 30l'$$

$$100 - l' = 3l'$$

$$l' = \frac{100}{4} = 25\text{cm}$$

$$40 - 25 = 15$$



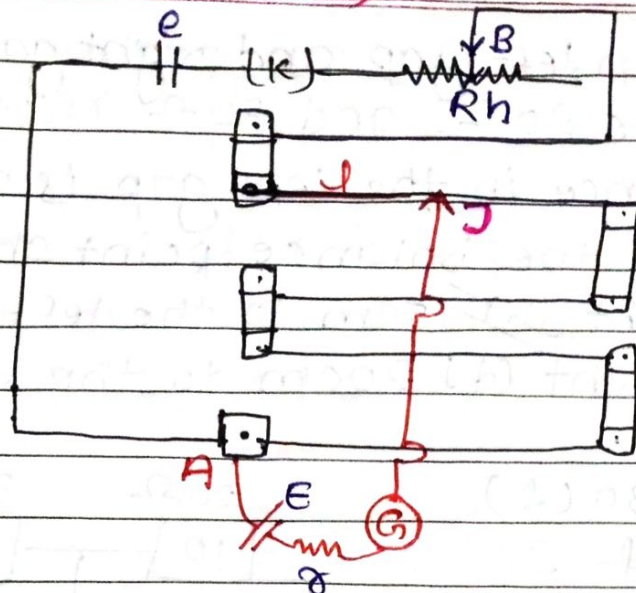
Potentiometer :- It is a device which is used for finding the internal resistance of a cell or comparing the emf of two cells.

Construction :- It consists of a number of wires having uniform length and cross-section area, stretched between two copper strips as shown in figure.

A cell of emf ($\mathcal{E}$) called driver cell connected with a point A with Rheostat (variable resistor). A jockey with Galvanometer ~~with~~ is connected to the point A. To find out the null point

(driver cell)

Date :



- Principle:- When constant current is passing through the potentiometer then potential difference across any section of wire is directly proportional to that section of wire.
- Theory:- We know that

$$V = iR$$

$$V = i \rho \frac{l}{A}$$

$$V = \left(\frac{i \rho}{A} \right) l$$

$$\boxed{V \propto l}$$

Application of potentiometer:-

(1) To comparing the emf of Two cell:-

To comparing the emf of two cell
Consider two cell of emf E_1 & E_2 are

Connected to point A with terminal 1 & 3 as shown in Figure.

Let when ① is connected with ② then null point founded by jockey at l_1 then principle of potentiometer.

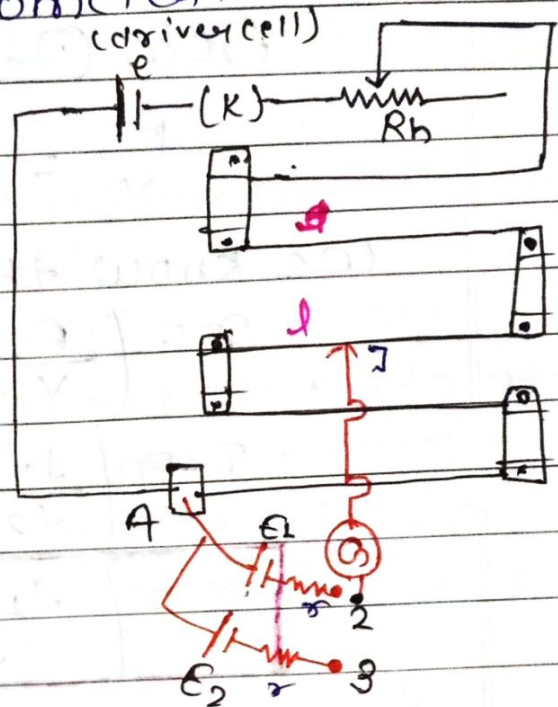
$$E_1 \propto l_1 \text{ --- (i)}$$

Similarly, when ③ is connected with ② then

$$E_2 \propto l_2 \text{ --- (ii)}$$

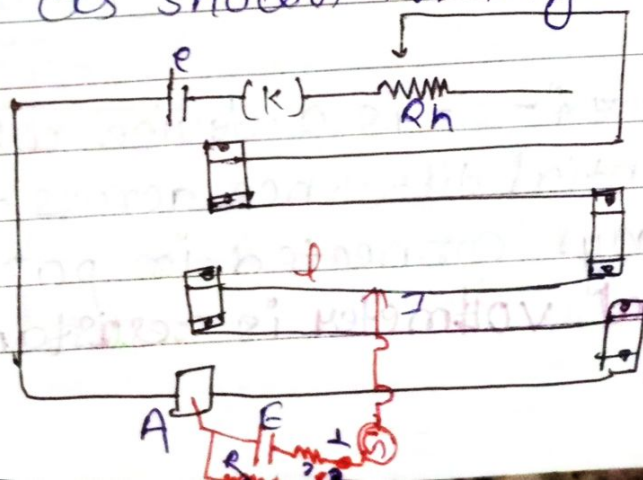
From (i) & (ii)

$$\frac{E_1}{E_2} = \frac{l_1}{l_2}$$



② To find out the internal resistance of a cell :-

Consider a cell of emf (E) of internal resistance (r) connected with terminal-1 also and external resistance (R) connected with terminal 2 in the parallel of cell as shown in Figure.



when 1 is not connected to terminal '2'
then null point founded by jockey at l_1 then

$$E \propto l_1 \quad \text{--- (i)}$$

Similarly, when (2) is connected with (1) then

$$V \propto l_2 \quad \text{--- (ii)}$$

From (i) & (ii)

$$\frac{E}{V} = \frac{l_1}{l_2}$$

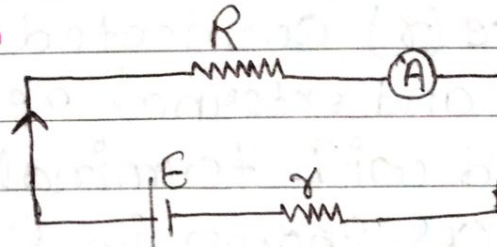
We know that

$$r = \left(\frac{E}{V} - 1 \right) R$$

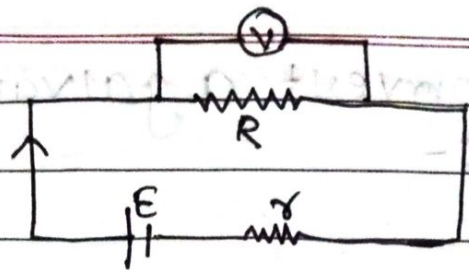
$$r = \left(\frac{l_1}{l_2} - 1 \right) R$$

$$r = \left(\frac{l_1 - l_2}{l_2} \right) R$$

*. Ammeter :- It is a device which measure the large current passing through the circuit and always connected in series.
For ideal ammeter its resistance must be low.



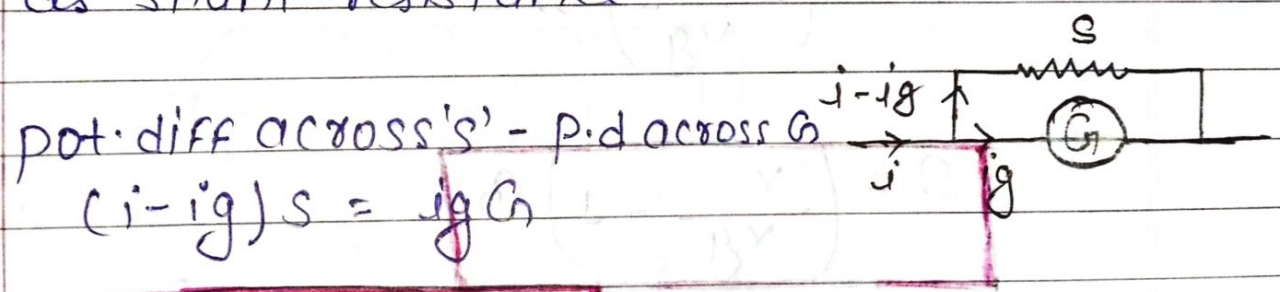
*. Voltmeter :- It is a device which measure the potential difference across the any circuit and always connected in parallel.
For ideal voltmeter its resistance must be large.



* Galvanometer:- It is a device which measure the small current passing through the circuit and it is more sensitive than the Ammeter.

How to convert a galvanometer into ammeter:-

To converting a galvanometer into ammeter a low resistance wire must be added to the parallel of galvanometer this wire is known as shunt and its resistance is known as shunt resistance.



$$S = \left(\frac{i_g}{i - i_g} \right) G$$

if n th part of main current passing through the galvanometer then

$$i_g = \frac{i}{n} \quad \left| \quad S = \left(\frac{\frac{i}{n}}{\frac{i}{n} - i} \right) G \right.$$

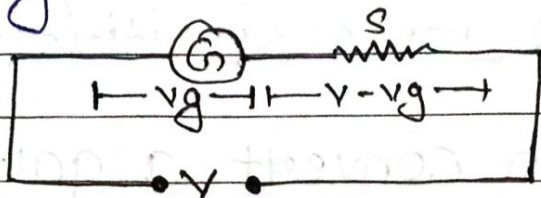
Now,

$$S = \left(\frac{\frac{i/n}{i - i/n}}{n} \right) G \quad \left| \quad S = \left(\frac{1}{n-1} \right) G \right.$$

#

How to convert a galvanometer into voltmeter :-

To converting a galvanometer into voltmeter a high resistance wire connected in the series of galvanometer



Current through G = Current through S .

$$\frac{V_g}{G} = \frac{V - V_g}{S}$$

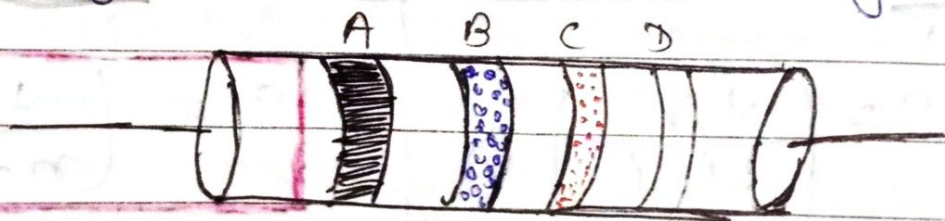
$$S \cdot V_g = (V - V_g) G$$

$$S = \left(\frac{V - V_g}{V_g} \right) G$$

$$S = \left(\frac{V}{V_g} - 1 \right) G$$

* Colour Code of Carbon resistance :-

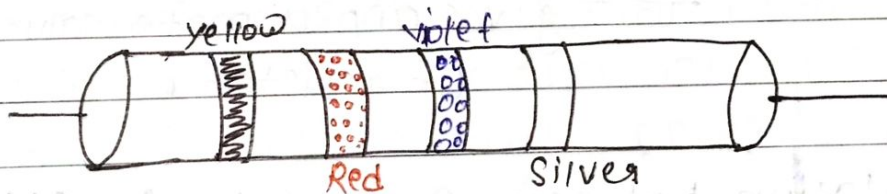
There are four colour bands A, B, C & D are marked on the carbon rod. In which first two bands A & B gives first two significant figure, third band C gives multiplier and fourth band D gives Tolerance



The codes different colours are given in the following table.

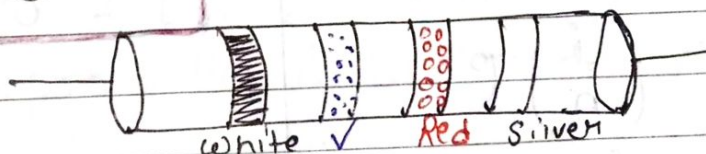
	Colour	Significant Figure	multiplier	Tolerance
1	Black	0	10^0	Gold = 5%.
2	Brown	1	10^1	Silver = 10%.
3	Red	2	10^2	NO-colour = 20%.
4	Orange	3	10^3	
5	Yellow	4	10^4	
6	Green	5	10^5	
7	Blue	6	10^6	
8	violet	7	10^7	
9	Grey	8	10^8	
10	White	9	10^9	

eg:-



$$R = 42 \times 10^7 \pm 10\%$$

Q1. In the figure a carbon resistor has bands of different colours on its body as mentioned in the figure. The value of the resistance is



(a) $2.2 \text{ k}\Omega$

(b) $3.3 \text{ k}\Omega$

(c) $5.6 \text{ k}\Omega$

(d) $9.1 \text{ k}\Omega$

$$R = 97 \times 10^2 \pm 10\%$$

$$= 9.7 \times 10^3 \pm 10\%$$

$$= (9.7 \text{ k} \pm 10\%) \Omega$$

*. Temperature dependence resistance:-

As we know that resistance is directly proportional to temperature.

Resistance at $t^{\circ}\text{C}$ is given by

$$R_t = R_0(1 + \alpha t)$$

where

α = Temperature co-efficient resistor

R_0 = Resistance at 0°C

If R_1 & R_2 are resistance of a conductor at temperature $t_1^{\circ}\text{C}$ & $t_2^{\circ}\text{C}$ then,

$$R_2 = R_1(1 + \alpha \Delta T) = R[1 + \alpha(t_2 - t_1)]$$

This gives approximate value of resistance.

*. Relation between Current density (j), Sigma (σ) & Electric Field (E) :-

We know that

$$V = iR$$

$$E \ell = j \rho \frac{\ell}{A}$$

$$E = \left(\frac{j}{A}\right) \rho$$

$$E = j \rho$$

$$\frac{E}{\rho} = j$$

$$\sigma E = j$$

$$j = \sigma E$$

* Relation between mobility (μ) & Conductivity:-
We know that

$$\mu = \frac{v_d}{E}$$

$$\mu E = \frac{1}{ne} \sigma E$$

$$\mu E = v_d \quad [j = nev_d A]$$

$$\mu E = \frac{j}{ne}$$

$$\sigma = \mu ne$$

$$\mu E = \frac{1}{ne} \left(\frac{j}{A} \right)$$

$$\mu E = \frac{1}{ne} j$$

* Thermal effect of current :-

(i) Joule's law of thermal effect of current :-
Due collision of free electron with lattice or +ve ion of conductor then loss of kinetic energy of free electrons appears on the surface of conductor in the form of heat
This Heat is :-

(a) Directly proportional to the square of the current,
i.e. $H \propto i^2$

(b) Directly proportional to Resistance

$$\text{i.e. } H \propto R$$

(c) Directly proportional to time,
i.e. $H \propto t$

Now combining all the above laws we get

$$H = i^2 R t$$

$$\text{or } H = k i^2 R t$$

But $k = 1$ in SI unit

Hence,

$$H = i^2 R t = v i t = \frac{v^2 t}{R}$$

Power :- The rate of doing work done by the source of emf is known as Power.

i.e.

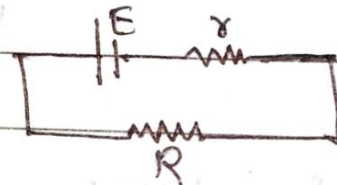
$$P = \frac{W}{t}$$

$$P = \frac{i^2 R t}{t}$$

$$P = i^2 R = v i = \frac{v^2}{R}$$

* Maximum power theorem :- It states that power will be maximum when internal resistance equal to external resistance.

$$\text{i.e. } r = R$$



maximum power is given by

$$P_{\text{max}} = \frac{E^2}{4r} \text{ or } \frac{E^2}{4R}$$